

BAHIRDAR INSTITUTE OF TECHNOLOGY
FACULTY OF MECHANICAL AND INDUSTRIAL ENGINEERING

1	School		School of Mechanical & Industrial Engineering			
2	Program		B.Sc. in Mechanical Engineering			
3	Course Title		Mechanisms of Machinery			
4	Course Number		MEng 3162			
5	ECTS		7			
6	Contact Hrs.		3 lecture, 3 tutorial, 1lab, 5 Home study (HS)			
7	Duration		1 Semester			
8	Number of Students Taking the Module		65 Students			
9	Course Owner(s)		School of Mechanical and Industrial Engineering			
10	Module Description	Introduction; Transmission of motion; Linkages; Kinematics analysis of linkages; Introduction to computer methods for kinematic analysis of linkages; cam design; Joints; Governors; Gear Trains; Introduction to synthesis; Force analysis of machinery; Engine torque fluctuation; Balancing of rotating and reciprocating masses; Gyroscopes.				
11	Prerequisite		Engineering Mechanics II (Dynamics)			
12	Learning Outcome	The Course enable the students to understand: The different types of linkage mechanisms used in mechanical design; The kinematic and kinetic analysis and design of machinery; Computer method for kinematic and kinetic analysis of mechanisms; Design and analysis of cams, universal joints, governors, gear trains, flywheels and gyroscopes; and Balancing of rotating and reciprocating machines.				
13	Module Content					
	Academic content of the module		Time allotted (Hr.)			
			Lecture	Tutorial	Laboratory	Self-Study
	Introduction to basic concepts		4	-	-	3
	Position Kinematic analysis of Linkages (1 weeks)		1	2	-	3
	Velocity analysis of linkages (1 weeks)		1	5	-	4
	Acceleration analysis of linkages (1 weeks)		1	5	-	4
	Introduction to Computer Methods for Kinematics Analysis of Linkages (1 weeks)		1	3	1	3
	Force Analysis of Machinery (2 weeks)		1	3	1	4
	Flywheels (1 weeks)		1	2	-	2
	Cam Design (1 weeks)		2	5	1	5
	Introduction to Synthesis of Machinery (1weeks)		0.5	1	-	5
	Universal Joints (1weeks)		1	2	1	2
	Gear Trains (1 weeks)		1	5	1	4
	Governors (1 weeks)		2	4	-	4
Balancing of Rotating and Reciprocating masses (1 weeks)		2	6	1	5	

	Gyroscopes (1 weeks)	1	5	1	4						
14	Learning activity and teaching methods	Lecture , Demonstration and Tutorial									
15	Learning resources / inputs and use of instructional technology	Books, power points and videos.									
16	Role of Instructor and Students										
	Instructor	Gives lecture, guides and moderates the students. He/She will help the students to clearly visualize problems and show methods to tackle them.									
	Students	Students will participate in lecture, but the students are expected to participate actively in the class during tutorial and lab hours by asking and solving problems.									
17	Assessment criteria and grading system	Assessment Type			Weight						
		Assignments			5 %						
		Quizzes			10%						
		Project			15%						
		Intermediate exam I			15 %						
		Intermediate exam II			15 %						
		Final Examination			50%						
		Grading System									
		As per the working senate legislation of AAU									
18	Modular requirement	Minimum of 75% attendance during both tutorial and lecture hours. 100 % Attendance during Laboratory and final examinations.									
19	Mapping of the course/module Outcome to the program Learning Outcomes	No	Course Learning Outcomes (CLO)		Programs Outcomes (POs)						
			1	2	3	4	5	6	7	8	9
	1	Analyzing engineering systems related to displacement, velocity, acceleration, and force in a mechanism using commercially available software like Working Model 2D, MSC ADAMS.	✓ 3				✓ 3				
	2	Design simple engineering systems using fundamental Principles, theories, and laws of engineering dynamic systems.			✓ 3						✓ 3
	3	Investigating various techniques of designing simple mechanisms to solve a given engineering problems.									✓ 3
	4	Develop scientific solutions and give interpretation to the result for simple engineering mechanisms.				✓ 2					
		1-Very little emphasis 2-Moderate emphasis 3-Strong emphasis									
20	Reading Materials										

	Text Book	Alem Bazezew (PhD), Mechanism of Machinery, Addis Ababa University Press, 2001
	References	Norton, Robert L., "Design of Machinery", WCB/McGraw-Hill, 1999. Meriam, J.L. and Kraige, L.G., "Engineering Mechanics- Dynamics", John Wiley and Sons, 1992. Shigley, J.E. and Uicker, J.J., "Theory of Machines and Mechanisms", McGraw-Hill Book Company, Inc., 1995. Khurmi, R.S and Gupta, J.K., "Theory of Machines", Eurasia Publishing House ltd., 1983. Arthur G. Erdman, George N. Sandor, Sridhar Kota, and Arthur G Erdman, Mechanism Design: Analysis and Synthesis (4th Edition), May 15, 2001. Erdman, Arthur G. and Sandor, George N., "Mechanism Design: Analysis and Synthesis", Prentice Hall International, Inc., 1997.